Model Predictive Control for Highway Decongestion and Safety

Joshua Stirling
Supervisor: Dr. Guilherme Fróes Silva

October 2026

Contents

1 Introduction **4**
- 1.1 Background . . . 4
- 1.2 Objectives and Scope . . . 4

2 Literature Review **4**
- 2.1 Introduction . . . 4
- 2.2 Modelling Techniques . . . 4
- 2.3 Shockwave Effect and Safety . . . 5
- 2.4 Model Predictive Control . . . 6
- 2.5 Model Predictive Control on Variable Speed Limit Signs . . . 6
- 2.6 Prediction Horizon . . . 6
- 2.7 Driver Non-Compliance Factor . . . 7
- 2.8 Model Predictive Control Cost Functions to Increase Safety . . . 7
- 2.9 Conclusion . . . 7

3 Project Objectives and Expected Outcomes **8**
- 3.1 Project Deliverables . . . 8
- 3.2 Risk Management and Ethics . . . 8
- 3.3 Quality and Sustainability . . . 9

4 Macroscopic Prediction Model **10**
- 4.1 METANET . . . 10
- 4.2 Mathematical Formulation . . . 11
- 4.3 Calibration . . . 12
- 4.4 Validation . . . 14
- 4.5 Discussion . . . 14

5 Model Predictive Control Design **15**
- 5.1 Conceptual Operation of MPC . . . 15
- 5.2 Integration with SUMO . . . 16
- 5.3 Optimisation Setup and Constraints . . . 16
- 5.4 Cost Function Formulation . . . 17
- 5.5 Solver Robustness and Verification . . . 19

6 Experimental Method **19**
- 6.1 Simulation Scenarios . . . 19
- 6.2 Performance Metrics . . . 20
- 6.3 Baselines and Statistical Protocol . . . 21
- 6.4 Assumptions . . . 21

7 Results and Discussion **22**
- 7.1 Merge Road: Standing Queue . . . 22
- 7.2 Straight Road: Cost Function Comparison . . . 24
- 7.3 Safety . . . 26
- 7.4 Realistic Sign, Driver Compliance and the Rule Benchmark . . . 26
- 7.5 Why the Controller Behaves This Way . . . 27
- 7.6 Discussion . . . 28

7.7 Limitations 29

8 Conclusion **30**

A Full Cost Function Results **35**

1 Introduction

1.1 Background

Highway decongestion and safety continues to be a popular topic of investigation, no doubt because traffic continues to take countless lives, money, and time every year. A large focus in recent decades has been on utilising digital traffic control methods that already exist on major roads, such as Variable Speed Limits (VSL) signs, in order to provide improvements in safety and congestion. This has been combined with more advanced control methods which predict future states of traffic rather than only responding, such as Model Predictive Control (MPC), in order to reduce traffic shockwaves. Shockwaves represent the rubber-band effect of traffic, which causes vehicles to rapidly decelerate and then accelerate. The main reason for the strong focus in reducing shockwaves in the literature is due to their clear link to higher travel times and increased risk of accidents. Using MPC on VSL has been found to be an effective method of reducing shockwaves, however the current literature makes it clear that more investigation into how best to set it up is required.

1.2 Objectives and Scope

The aim of this research project was to find which MPC cost functions are most effective at reducing traffic congestion and increasing safety, in order to determine how to best implement MPC for optimal real-life benefits. This involved setting up the simulation environment, MPC controller, and testing various cost functions. The scope of this project was limited to two traffic scenarios: a lane-merge bottleneck (to produce a standing queue) and a straight two-lane road with a temporary slow-vehicle disturbance (to produce a travelling shockwave). This project did not explore which simulation or types of MPC methods are most optimal. The use of connected vehicles was also not considered in this project.

2 Literature Review

2.1 Introduction

The area of MPC for Highway Decongestion and Safety has had a large presence in the literature for traffic control due to the huge benefit of being able to predict future states of traffic and react before congestion is visible. In recent years, digital control inputs have been an increasingly popular method of applying MPC, as they already exist on most major roads. However, limited exploration has been conducted into which MPC cost function is best at reducing congestion and increasing safety. Thus, the aim of this literature review is to explore current implementations of MPC, how they benefit traffic in terms of congestion and safety, and how key parameters of MPC are tuned.

2.2 Modelling Techniques

The popularity of traffic modelling means there are many different techniques and software packages available to use. These models are usually either microscopic or macroscopic, with some hybrid versions existing too. Microscopic models are more accurate and computationally heavy because they simulate individual vehicles and their specific vehicle

behaviours. Macroscopic models have higher computational efficiency as they simulate the overall flow of traffic by assuming it to be a continuous fluid. However due to this assumption, they are less accurate to real-world conditions.

METANET is the macroscopic model most used by researchers [1–4]. METANET is a set of nonlinear difference equations for traffic density, speed, and flow developed in the 1990s, and is accessible through most major platforms, such as MATLAB and Python. It is extremely popular, can simulate traffic shockwaves, and allows for efficient testing of control techniques. However, a microscopic model such as SUMO provides far more accurate traffic simulations, as it simulates discrete vehicles and accounts for their interactions [5]. SUMO is an open source and well documented software package, and allows for real world data to be used in simulations. It can provide high detail analytics, which gives it a distinct advantage over the averaged and simplified macroscopic analytics provided through a platform such as METANET. The greatest drawback to using SUMO is that as a microscopic model, it requires far more computational power and time for its simulations, and is thus not ideal for optimisation models that require numerous simulations to be done.

A viable method for testing control techniques which leverages the speed of macroscopic models and the accuracy of microscopic has been demonstrated by Hadiuzzaman et al. [3] and Yang et al. [6]. They used a macroscopic model for control design, and then a microscopic model as a 'ground truth' to validate the results. As METANET is one of the more complicated macroscopic models, simpler models such as the Lighthill-Whitthham-Richards (LWR) model could be used. LWR is a first order model, and represents traffic through a single conservation equation, and is thus easier to implement for initial testing, though less accurate [7].

2.3 Shockwave Effect and Safety

Mitigating shockwaves has been the focus of a large portion of the work done within the traffic control domain [1, 8–11]. Shockwaves are the phenomenon produced by sudden changes in vehicle velocity, which travel down a stream of traffic until they reach the end or dissipate. This occurs when drivers are forced to brake suddenly, as the vehicles behind them brake too, causing the flow on effect. This increases the chances of collision dramatically, as was confirmed by Abdel-Aty et al. [12]. The shockwave effect has also been found to be linked to increased fuel consumption by Wu et al. [13], due to the energy wasted by accelerating and then braking. The adverse effects of shockwaves appear to be unanimous across all the papers surveyed, including credible sources such as Transportation Research Part B and Part C. The main limitation of these papers is that they rely heavily on simulation, however the adverse effects of traffic shockwaves has also been documented through real life data analysis.

Minimising the variance in vehicle speed has been shown to directly reduce the formation of traffic shockwaves and their subsequent adverse effects by multiple researchers [8–10, 12, 14]. This has led to various attempts to apply control law in order to reduce these shockwaves, reducing velocity variance (among other objectives) through strategies such as Model Predictive Control (MPC).

2.4 Model Predictive Control

The current review surveys of traffic control techniques regard MPC as an extremely promising control method for traffic [2, 15]. MPC predicts future system behaviour by estimating a model of the system, and then solves an optimisation problem to determine the best next possible control input sequence by minimising a cost function with set objectives.

Siri et al. [15] highlights that MPC's ability to predict traffic rather than only responding, as well as its ability to handle multiple chosen objectives, makes it extremely useful and versatile. They found through their survey that MPC consistently outperforms reactive control based techniques. Delle Monache et al. [2] confirm this, explaining similar benefits and even stating that MPC is part of the 'new generation' of control methods. They also highlighted the flexibility of MPC being used across different scenarios, with varied control inputs and objectives.

2.5 Model Predictive Control on Variable Speed Limit Signs

Further investigation has been done into applying MPC on Variable Speed Limit (VSL) signs and evaluating if it is an effective control input method. Due to VSL allowing for real time control over large portions of highway, and its existing presence on most highways, it has been a popular choice of applying MPC.

Hegyi et al. [4] applied MPC to VSL using METANET for a highway merging bottleneck scenario. They found that MPC was able to significantly reduce the size and severity of shockwaves formed. In a no control case, the shockwaves would propagate down the entire stream of traffic, however with MPC applied they dissipated completely after approximately 90 minutes. This resulted in an average Travel Time Spent (TTS) reduction of approximately 20%. Their paper presented strong findings, however they did not validate them using a microscopic model or real world data, and they assumed full driver compliance in all simulations.

2.6 Prediction Horizon

Deng and Zhou [1] conducted a more recent study into applying MPC on VSL using METANET. Their objective was to reduce TTS, but also to find the optimal prediction horizon length, which is a crucial variable in MPC as it dictates how far forward the system's dynamics are considered. They found their best TTS improvement of 13% with a prediction horizon of 60 minutes, however what was more insightful was the trends they found, as they can be applied to other traffic scenarios. They found that prediction horizons lower than the optimal value performed exponentially worse, while prediction horizons higher than the optimal value decreased in effectiveness linearly. Shang et al. [11] found similar results, stating that too short of a prediction horizon limits the controller's foresight, while too long of a prediction horizon increases computational difficulty and makes it harder for the controller to converge to an optimal solution.

A strength of Deng and Zhou's [1] paper was that they explored factors such as Total Fuel Consumption (TFC) and Total Pollutants Emitted (TPE) in their analysis. With an optimal setup, they found pollutants to be reduced by approximately 37%, and fuel consumed to be reduced by over 20%. A limitation of both papers is that they present

no quantifiable method to find an optimal prediction horizon length, and only provide overall trends. Furthermore, similar to Hegyi et al. [4], they rely on simulations without any concrete validation, and they assume full driver compliance.

2.7 Driver Non-Compliance Factor

The need for a realistic driver non-compliance factor has been highlighted by numerous studies [12, 14], as most papers inaccurately assume 100% compliance [1, 4, 8–10]. The driver non-compliance factor makes simulations more realistic by mathematically including the real-life factor of drivers ignoring speed limits and travelling at a speed either slightly higher or lower. Yu and Abdel-Aty [14] found that with high driver compliance, the benefits of control law on a system were strong, at moderate compliance the benefits dropped noticeably, and at low compliance the benefits were barely noticeable.

A strength of Yu and Abdel-Aty's [14] study was their explicit focus on the driver compliance factor, which most studies ignore. However a limitation is that they restricted the factor to a set constant, and didn't explore compliance methods that may be more accurate to real life conditions. These could include altering the compliance according to location and conditions, or choosing the compliance for each vehicle from a random probability distribution.

2.8 Model Predictive Control Cost Functions to Increase Safety

Out of the work done in applying MPC to VSL, minimal investigation has been done in regards to which specific cost functions can best reduce shockwave formation in traffic. Hadiuzzaman et al. [3] simulated and tested delay and velocity variance based objectives, and found that when each objective was applied singularly, it best reduced its target metric, and was marginally helpful in improving the other objective. However, they found that when both objectives were applied, both metrics improved substantially, although not as much as when they were focused on individually. Yang et al. [6] did a similar investigation, however they validated the results using the microscopic model VISSIM instead of purely using a macroscopic model. They found similar trends, and suggested that "Future work should examine different formulations of the objective function and their impacts on both mobility and safety performance." Both of the studies only considered the combination of two objective components, and didn't include any broad exploration into other cost function formats.

2.9 Conclusion

Model Predictive Control, due to its ability to predict future events, has been shown to be one of the most promising highway traffic control techniques. It has been tested mainly on Variable Speed Limit systems through macroscopic models, which has confirmed that it is capable of reducing shockwave formation, travel time, pollutants emitted, and fuel consumption.

From key review papers and the current literature as a whole, it is clear that the ability to reduce shockwaves significantly improves safety and congestion. Several studies have used MPC for this objective and have confirmed its potential use, as well as clarified how MPC

should be setup and tuned. Deng and Zhou [1] found that as general trend, the prediction horizon length should be set at the furthermost point at which an accurate estimation can be made. Although almost all studies assumed full driver compliance, Yu and Abdel-Aty [14] found that although control effects weakened, they were still effective with moderate compliance. Hadiuzzaman et al. [3] and Yang et al. [6] found that combining the MPC cost function objectives of travel time and velocity smoothing gave results which gave substantial improvements to both metrics, leading to decreased congestion and increased safety.

The work which has already been conducted strongly suggests that applying MPC on VSL has numerous benefits, however the research done into which specific cost functions can achieve these benefits is minimal. There is limited research into which conditions should be used and how they should be weighted, other than travel time and velocity smoothing specifically. There is also further need for more accurate validation of results from macroscopic models, using real life data or microscopic simulations.

3 Project Objectives and Expected Outcomes

3.1 Project Deliverables

1. Complete project report: a complete written report documenting literature review, methodology, simulation design, results, and analysis of MPC cost function impacts on highway safety and traffic flow.

2. Simulation models: two configured SUMO highway models, Bounded-METANET prediction model, and non-linear MPC for use with a library of 30+ cost functions.

3. Data and analysis outputs: plots, graphs and processed metrics for each cost function (across varying highway scenarios if applicable).

4. Oral presentation: summary of objectives, methodology, findings and contributions, and suggested future areas of work.

All deliverables have been completed (through this report, the oral presentation, and upload of all files to GIT), and progress has been reported through regular meetings with the project supervisor throughout Semester 2 in 2025 and 2026.

3.2 Risk Management and Ethics

Due to the project methods being applied purely through simulation, there were no immediate physical risks to the researcher or anyone else directly or indirectly involved. However, it was identified that there would be potential risks if the project results were used in real life implementation, which could occur due to incorrect use or a fault in the findings. These risks could range from delays in traffic, to increased risk of crashes and fatalities. Thus, great care was taken in presenting accurate findings, in a clearly communicated manner.

This research project has followed the QUT guidelines for responsible research in order to ensure academic integrity. As the project methods only interfaced with simulation

platforms such as SUMO, there were no restrictions on the methods applied, however the work done was bound by academic standards and ethical reporting. All modelling and analysis was carried out in accordance to the Engineers Australia Code of Ethics, particularly in regards to integrity, competency and public good. This ensures that results were reported transparently, limitations of the model were acknowledged, and no conclusions were misrepresented. Thus, contradictions, negative results, and weaker findings have been clearly displayed as such in this report.

The main constraints of the project were due to having limited time, and a lack of industry resources. The project was required to be finished after two semesters, and the institution the project was being completed through (QUT) does not have access to the most advanced traffic simulation tools, such as PTV VISSIM or AIMSUN Next, which require expensive licenses. The simulations were completed through SUMO on a local device, causing a full cost function search to require 24 hours of uninterrupted time.

3.3 Quality and Sustainability

The quality of project findings and documentation have been upheld by adhering to the guidelines and marking criteria set by Engineers Australia, QUT Faculty of Engineering, and through regular consultation with the project supervisor. Each simulation configuration was tested over 10 random seeds to improve the reliability of the findings, and cost functions were compared over the same seeds to ensure fairness in comparison. The validity of the results was checked by comparing the trends with those reported in previous research, and by validating metric results against individual vehicle trajectories in the simulation. Extensive verification of the model and controller was also completed.

The findings of this report, if accurate and implemented correctly, could have a direct impact to the levels of pollutants produced by vehicles and fossil fuels used in the locations where it is applied. As the existing literature has suggested, MPC is capable of reducing fuel consumption and pollutants emitted, because it influences the flow of traffic to be more stable, reducing braking and acceleration. Thus this project has the potential to aid in environmental sustainability. However, CO_2 analysis through SUMO could not find a statistically significant reduction, and thus sustainability improvements cannot be supported by the evidence at this point.

The findings and proposed MPC implementations will have a long term impact. This is because MPC by design is extremely adaptable, and the implementations used can be used in drastically different conditions, with the appropriate parameter tuning. The aim of this report was not just to solve a specific scenario, but to give a broader understanding of different cost functions and their impact, and if possible, which cost functions are best for specific scenarios. The infrastructure required to implement this research is unlikely to be removed, and it is likely that more control inputs will be available to highways in future, through the advancement of Connected Autonomous Vehicles (CAVs). Therefore, the findings of this project have a high likelihood of remaining relevant for the foreseeable future.

4 Macroscopic Prediction Model

SUMO is a microscopic simulation environment which simulates the behaviour of every individual vehicle. It is computationally impractical to use SUMO directly with a controller such as MPC, which needs to run a model numerous times in order to find the next best control inputs. In order to bridge the gap, a macroscopic model can be used in the MPC controller to drastically improve computational efficiency. The goal of the macroscopic model is to capture the overall flow of the traffic in a way which matches the SUMO model, which can be done orders of magnitude faster than simulating individual particles of traffic. The macroscopic model was updated with live SUMO data whenever MPC was run so that the model did not 'drift' away from the ground truth. This meant that an exact replica was not required.

4.1 METANET

Initially, a first-order macroscopic model, LWR [16], was tested. However, it was rejected as it computes speed directly from density, making the testing of cost functions built on speed changes irrelevant to the model. Thus METANET [4, 17] was chosen as it is a second-order model and the standard implementation for VSL MPC.

The standard implementation of METANET updates a segment's speed using three terms:

1. Relaxation, which is the traffic drifting towards its ideal speed for its current density.
2. Convection, which is the speed carried from the upstream segment.
3. Anticipation, which is traffic slowing down in proportion to how dense the next segment is.

Testing was done by running the same starting and flow conditions through SUMO and METANET for the merge-road scenario, without control. Through testing it was found that the anticipation term caused METANET to lose performance as a predictor for the given situation. When anticipation was weak, vehicles didn't slow down in preparation for downstream density, which caused densities to rise beyond the physical capabilities of the model. When anticipation was too strong the model forcibly removed separation between queues and free-flowing traffic, blurring the speed across all segments. This blurring can be seen in Figure 1, where the standard implementation of METANET can be seen to create a uniform band across all road segments, whereas SUMO and Bounded-METANET show the queue forming and releasing. A middle ground between the two extremes was attempted to be found, however although the results improved with tuning, bounded METANET was found to be a better implementation.

Bounded METANET [18] keeps only the relaxation term from the aforementioned list. It then adds a look-ahead density, which is where drivers aim for the optimal speed based on a weighted average between the current and next segment. This implementation is called bounded because each new speed is a weighted average of the current speed and a target speed between zero and the free-flow speed, thus ensuring it always stays between the two. This was implemented using the sym-metanet library [19] on the CasADi engine [20], as it has been verified as a reliable and proven implementation of METANET.

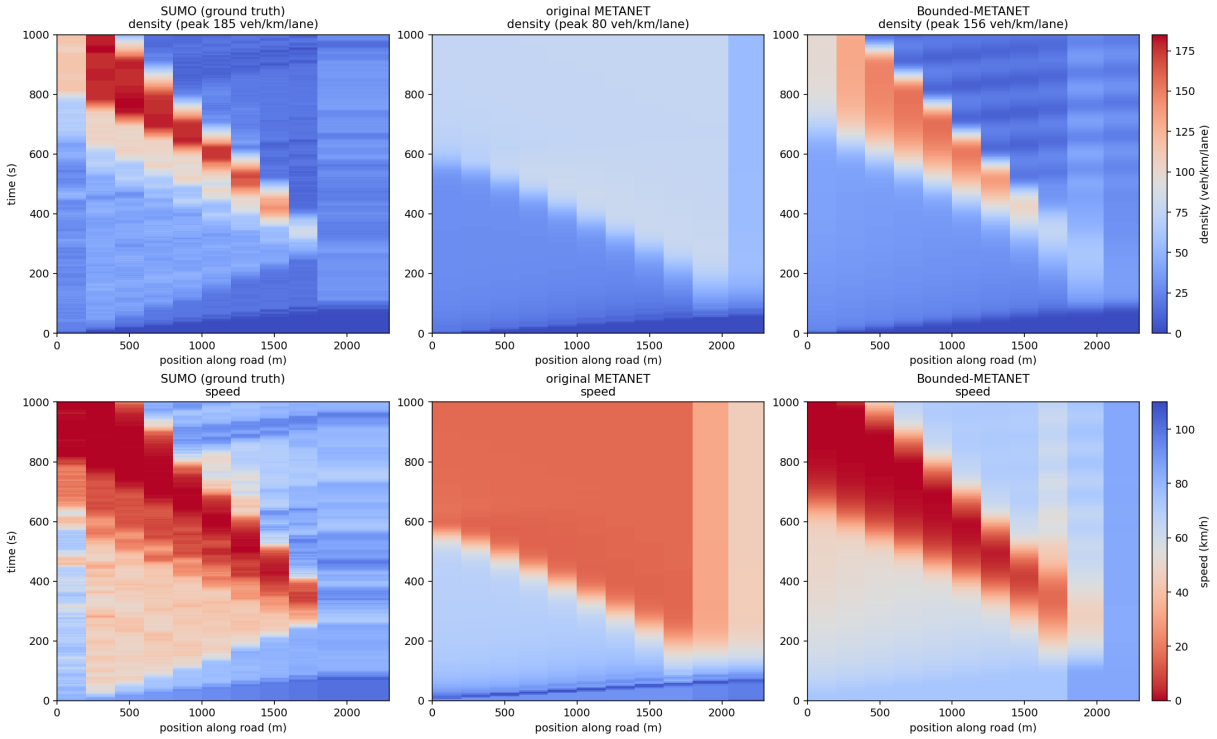


Figure 1: Macroscopic Model Comparison: Density (top) and speed (bottom); SUMO (left), standard METANET (centre), calibrated Bounded-METANET (right)

4.2 Mathematical Formulation

In the implementation, the road is first divided into 0.2 km segments, to match the SUMO edges. The model updates every $T = 5$ s, which satisfies the stability condition $Tv_f < L_i$, which requires traffic to not be able to cross more than a single segment in one time-step. Each segment i has a density ρ_i (veh/km/lane) and a speed v_i (km/h), with time steps being denoted as k. Speed is its own state (unlike LWR), and density follows the conservation of vehicles,

$$\rho_i(k+1) = \rho_i(k) + \frac{T}{L_i \lambda_i}\big[\, q_{i-1}(k) - q_i(k) \,\big], \qquad q_i(k) = \lambda_i\, \rho_i(k)\, v_i(k), \tag{1}$$

where λ_i is the number of lanes and q_i is the flow.

As mentioned, Bounded-METANET does not include the convection and anticipation terms which METANET has, and keeps only relaxation towards a target speed,

$$v_i(k+1) = v_i(k) + \frac{T}{\tau}\big[\, V\big(\hat{\rho}_i(k)\big) - v_i(k) \,\big], \tag{2}$$

where τ is the relaxation time and $V(\rho) = v_f \exp\big[-\frac{1}{a}(\rho/\rho_{cr})^a\big]$ is the exponential fundamental diagram, v_f is the free flow speed, ρ_{cr} is the critical density, and a is the shape parameter. Anticipation of future sections is captured through a look-ahead density,

$$\hat{\rho}_i(k) = \rho_i(k) + \tilde{\eta}\, \frac{\tilde{\kappa}}{\rho_i(k) + \tilde{\kappa}}\big[\, \rho_{i+1}(k) - \rho_i(k) \,\big], \tag{3}$$

which is a weighted average of the current and next segment's densities. This term is what causes the drivers to begin slowing down before they reach denser traffic. The main

distinction between look-ahead density and standard METANET's anticipation term is that because T/τ lies between zero and one, each new speed is a weighted average of the current speed and a target between zero and v_f. Thus, predicted speeds can never become negative or exceed free flow [18].

On segments with VSL the target speed is capped by the posted limit u_i:

$$V_i^e(k) = \min\left\{ V\big(\hat{\rho}_i(k)\big),\ \frac{u_i(k)}{u_i^{\text{legal}}}\, v_f \right\}. \tag{4}$$

This equation ensures that the segment cannot flow faster than it would naturally be able to for the defined density. The METANET model was found during tuning to operate best when it assumed that the maximum speed the cars can travel was lower than the true value. Thus the inputs to the model need to be scaled to the model's version of the maximum speed limit, and the posted limit is expressed as a fraction of the legal limit multiplied by the free-flow speed.
Inflow to the first road segment is described as

$$q_0(k) = \min\left\{ d(k),\ C_o,\ C_o\, \frac{\rho_{\max} - \rho_1(k)}{\rho_{\max} - \rho_{cr}} \right\}, \tag{5}$$

where $d(k)$ is the flow measured in SUMO just upstream of the first sign over the last control interval, held constant across the horizon, and C_o is the entrance capacity. The model's entrance queue is reset to zero at each solve, so vehicles queued further upstream are not part of the prediction. The third term in equation 5 reduces inflow as the first segment approaches its maximum density $\rho_{\max}$. To allow vehicles to freely flow out of the simulation the road beyond the modelled section is treated as a free exit.

4.3 Calibration

To calibrate the METANET model, several results were compared with an uncontrolled SUMO simulation starting at the same state. This was done using a score which ranked parameter selection based on how well the model reproduced the whole simulation and how accurately it predicted traffic 210 s into the future. Parameter sets that produced physically implausible behaviour were also penalised.

For the straight road the length of the jam and the time frame it took to clear were also considered in its scoring. Two different vehicle inflow rates (4,000 and 4,600 veh/h) were considered for the testing. The calibrated parameters can be seen in Table 1. On the test runs the calibrated model had a jam discharge of 4687 veh/h compared to 4700 veh/h in SUMO, and cleared the released jams in 400, 375 and 340 s against SUMO's 468, 408 and 348 s (across three different tests, respectively). Figure 2 shows one of the speed and jam density plots for one of the test runs. The model can be seen to reproduce the jam which travels upstream, over a similar period, and to a similar severity. However, the jam can be seen to clear faster in the macroscopic model.

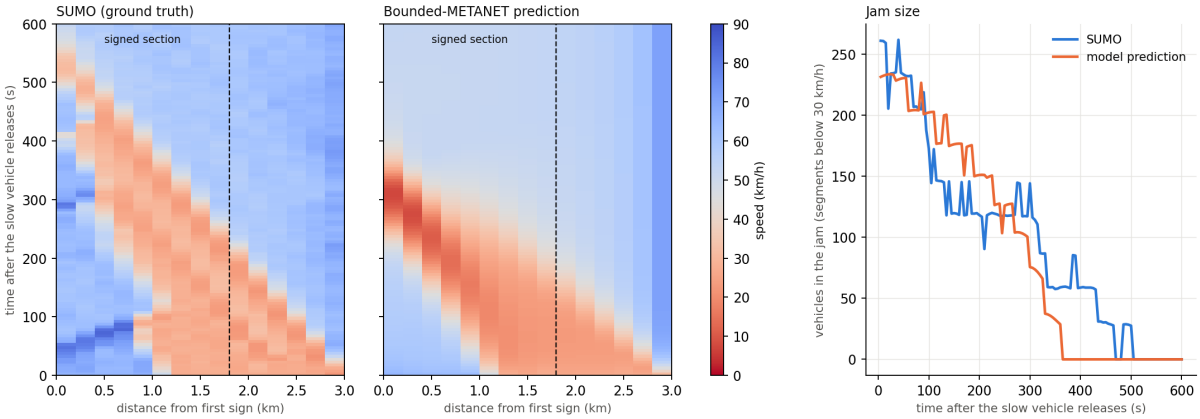


Figure 2: Straight road (4600 veh/h) after the jam releases: SUMO speed (left), model prediction from the same state (centre), and vehicles in the jam (right).

Table 1: Calibrated Bounded-METANET parameters for each road

Parameter	Description	Merge road	Straight road
v_f (km/h)	Free-flow speed	74.0 (85.2 one lane)	71.3
ρ_{cr} (veh/km/lane)	Critical density	35.4	47.3
a	Fundamental diagram shape	2.89	2.65
τ (s)	Relaxation time	48.8	24.1
$\tilde{\eta}$	Look-ahead weight	0.70	0.96
$\tilde{\kappa}$ (veh/km/lane)	Look-ahead smoothing	1733	296
$\rho_{\max}$ (veh/km/lane)	Maximum density	98.3	118.7
C_o (veh/h)	Entrance capacity	4075	4554

4.4 Validation

The prediction model was validated by comparing its predictions to a baseline prediction for 600 seconds, which was deemed long enough to capture the formation and clearing of a jam, whilst also exceeding the controller's longest planning look-ahead of 420 s. The baseline used, called persistence, involved predicting that nothing would change over the 600 seconds. Uncontrolled SUMO simulations were run for each road (three seeds at each of 4600 and 4800 veh/h for the straight road, and three seeds for the merge road). Every 30 seconds, the model was started using SUMO's measured densities and speeds, and its predicted speeds were then compared with SUMO's at each look-ahead time. For the straight road scenario, only predictions which included a jam within their time frame were included in the results.

Table 2: Speed prediction error (RMSE, km/h) of the calibrated model and of persistence, by look-ahead time

	Straight road		Merge road	
Look-ahead (s)	Model	Persistence	Model	Persistence
30	7.2	7.1	8.6	10.4
60	9.1	11.0	12.5	16.5
120	11.5	15.7	18.6	23.6
210	14.7	20.3	26.5	29.7
420	16.0	25.0	34.8	32.0
600	13.1	23.4	35.6	30.5

Table 2 shows that on the straight-road scenario, the model performs similarly to the baseline with a 30-second look-ahead but then vastly outperforms it at every longer look-ahead tested up to 600 seconds, with approximately 28–44% less error from 210 s onwards. In the merge road scenario the model outperforms the baseline up to 210 seconds, but is beaten at 420 seconds and beyond. Although this would appear that the model is worse at predicting for the full look-ahead time, the merge scenario has a large stationary queue, meaning that persistence is already a strong prediction method for this scenario over long time periods, as a queue that does not move changes very little.

4.5 Discussion

The calibrated Bounded-METANET model was deemed to be a good enough foundation for testing, as it ensures realistic speeds (through being bounded), which in turn smooths the equations for the optimiser. On the straight road scenario, it predicted traffic more accurately than persistence at every look-ahead beyond 30 s, and it correctly reproduced the clearing of the jam, with similar discharge rates and clearance times.

A limitation identified about the model is that in order to match the model's performance with SUMO, the maximum density was set at approximately half to two-thirds (98–119 compared to 182 veh/km/lane in SUMO). This is a known limitation of models which assume single-value relationship between speed and density (one speed per density), as in reality the speed can be different for the same density of traffic, depending on the behaviour of the queue (whether it is slowing down, speeding up etc) [16]. This

was also identified to potentially be another large factor behind why the merge scenario was predicted less accurately (due to it having a longer queue to be influenced by this simplification). A further limitation is that the model's strength is in predicting, which it cannot do in the straight-road scenario as the disturbance is added in. However, what was identified to be the main limitation was that the model was tuned on a no-control scenario and not tuned for its effectiveness in predicting the affects of control action (this is further explored in Section 7.5).

5 Model Predictive Control Design

MPC was chosen as the control method for this investigation because of its reported advantage over traditional reactionary control methods. This advantage has been expressed in the plethora of research conducted on using MPC for control in traffic, as well as its popularity of use in many other areas.

The goal of MPC in this investigation is to minimise congestion and improve safety by applying control actions that prevent shockwaves.

5.1 Conceptual Operation of MPC

MPC predicts future system behaviour using a model (such as the Bounded-METANET model of Section 4), and then solves an optimisation problem that minimises a cost function with chosen objectives. It does this by simulating how the system will evolve under various control sequences to see the effect on states such as density and speed. The expression of how the system states evolve is represented as

$$x_{k+1} = f(x_k, u_k, d_k), \tag{6}$$

where x_k represents the system state (the density ρ and mean speed v of each segment) at time step k, u_k represents the control input (the posted speed limits), d_k is the forecast inflow entering the modelled road, and $f(\cdot)$ denotes the prediction model used to forecast how the system evolves (Bounded-METANET, Section 4).

This information is then used in an optimisation problem to find which actions achieve the result that minimises the cost function. The horizon of the MPC controller refers to how far in the future it simulates. Shorter horizons take less of the future effects into account, however longer horizons require the model used to be more accurate. The cost function penalises objectives based on their weighting, and these factors can be altered in order to produce the desired response from the controller. The first control action from the optimal solution found is applied, and then the process is repeated after a set length of time. The chosen prediction horizon time was 420 seconds into the future.

The freedom of choice regarding the cost function is what allows MPC to be so versatile and adaptable, but also presents the challenge identified previously in the literature review.

5.2 Integration with SUMO

The purpose of the predictive framework described in the previous section is to find the optimal speed limit suggestions for each section. The prediction is made using the live densities and speeds per 200 m segments from the SUMO simulation, and the speed limits are then used to update the SUMO simulation for a set time period. This allows for the effects of the updated control law to influence the microscopic model, before the process is then repeated again. Thus running MPC alongside the SUMO simulation essentially allows for a lightweight way to look into future possibilities and choose the outcome which appears best. If the MPC controller is configured to minimise factors which lead to shock-waves and congestion, then theoretically this model will provide the control inputs which best achieve that goal.

The MPC model was integrated with SUMO using the Traffic Control Interface (TraCI) API. This allows for real time communication between the controller built in Python, and the SUMO simulation environment. At each control cycle, the controller receives the density and speed of each section, and sends back the suggested section speed limits. These are then applied at the start of the next simulation step. The controller only switches on when traffic is measured as present. In both scenarios, a 3.6km stretch of uncontrolled road was added before the controlled segment, to give visibility in the results of the effects the controller causes upstream of its control actions (such as queuing traffic), although the prediction model itself only covers the road from the first sign.

5.3 Optimisation Setup and Constraints

As the traffic model is non-linear, the controller was also designed to be nonlinear. The Bounded-METANET equations of Section 4.2 are included directly as constraints in the optimisation problem, so the prediction is exact with respect to the model. The problem was built with CasADi [20] and the csnlp library [21], which provide the exact derivatives of the model, and solved with the interior-point solver IPOPT [22].

At each control interval, the controller chooses the next N_c speed limits for each of the nine signs by solving

$$\min_{\mathbf{u}_1,\ldots,\mathbf{u}_{N_c}} J(\mathbf{x}, \mathbf{u}), \tag{7}$$

where $\mathbf{u}_m$ is the set of nine displayed speed limits for control move m and J is the cost function (Section 5.4). Through the choice of design, the optimisation is bound to four constraints:

- **Traffic model**: The predicted traffic must follow Bounded-METANET (Section 4.2).
- **Speed-limit range**: Every limit lies between 40 km/h and the legal limit.
- **Rate limit**: No sign changes by more than $\Delta u_{\max} = 60$ km/h across one move (60 seconds).
- **Physical states**: Predicted densities, speeds and the entrance queue cannot be negative.

The effect of the limits is observed through applying them to the Bounded-METANET model, and then by judging the predicted effects of traffic they would cause. The posted limits were constrained to a minimum of 40 km/h, taken as the lowest limit a sign would reasonably display on a highway. The rate limit was kept wide, so that the model could respond quickly to disturbances and the formation of queues, however in practice it never bound, as the allowable range of every sign is less than 60 km/h.

The realistic controller was designed to re-plan every 60 seconds, meaning a single control move is the same length. However, the model looks 420 seconds ahead during its prediction (seven 60 s moves), and only applies the first control move. This allows for the model to run at a higher frequency while accounting for longer term effects. Speed limits are rounded to the nearest multiple of 10 km/h to ensure realism. A consequence of this is that the sings cannot display the 84.6 km/h legal limit while control is active (being rounded to 80 km/h), harming TTS slightly. An idealised controller, which runs every 30 seconds and has no rounding was also tested, and its results reported in Section 7.4.

5.4 Cost Function Formulation

Each cost function used in the controller is a weighted sum of terms,

$$J = \sum_j w_j J_j. \tag{8}$$

and is applied over the full predicted traffic for the entire length of the horizon. Almost the entirety of the cost function sets which were found to be effective were pairs, including total time spent (TTS) with a weight of one, as one of the terms. This meant that these cost functions traded a single other traffic goal with mobility. TTS counts the vehicle hours spent on the road and in the queue [4]:

$$J_{\text{TTS}} = T \sum_k \Big(\sum_i \rho_{i,k} L_i \lambda_i + w_k \Big), \tag{9}$$

where $\rho_{i,k} L_i \lambda_i$ is the number of vehicles in segment i at step k and w_k is the entrance queue. It was found necessary to include the queuing term in TTS, as without it the controller would hold vehicles back at the entrance, where they are no longer counted on the road, and thus reduce TTS.
Another important term was a count of the vehicles moving below 30 km/h, which was used as an indication of vehicles within a traffic jam:

$$J_{\text{jam}} = T \sum_k \sum_i \rho_{i,k} L_i \lambda_i \cdot \tfrac{1}{2} \Big[1 + \tanh \Big(\frac{30 - v_{i,k}}{3} \Big) \Big]. \tag{10}$$

The tanh factor was used to smooth the cut-off, rather than having a hard cut at 30 km/h. This was done so that IPOPT had a gradient to follow. All the other terms considered are listed in Table 3. Several terms were designed for this project, and are thus without a source.

As different terms can be measured in units which have vastly different scales, each term was scaled so that it would have as much influence on the speed limit as TTS does (TTS being used as the baseline, as it was the most common term). This was done

Table 3: Cost function terms.

Term	Penalises	Source
Mobility		
Total time spent	Vehicle-hours on the road and in the entrance queue	[4]
Delay	Time spent beyond the free-flow journey time	[23]
Terminal	Vehicles still on the road at the end of the horizon	–
Entrance queue	Squared entrance queue	–
Shockwave		
Jam	Vehicles moving below 30 km/h	–
Acceleration	Squared acceleration, weighted by the vehicles experiencing it	[24]
Temporal acceleration	Only the speed changes within a segment (stop-and-go)	[24]
Density-weighted acceleration	Squared acceleration, weighted by density	–
Density rate	How quickly density rises in a segment (a sharp jam front)	–
Upstream congestion	Density above critical, weighted more heavily further upstream	–
Squared density	Squared density in every segment	[25]
Density tracking	Deviation from a target density near critical (preliminary design)	–
Safety		
Speed variance	Spread of speeds along the road	[3, 6]
Speed drop	Squared drop in speed between adjacent segments	–
Conflict risk	Speed drop multiplied by the density approaching it	–
Sign behaviour		
Control variation	Squared change in the posted limits between moves	[4]
Merge road only		
Bottleneck density	Density above critical in the merge segment	–
Bottleneck discharge	Low flow out of the merge (rewarded earlier in the horizon)	–

through running the model 420 s ahead and comparing the influence of altering each term compared to TTS. Altogether, using standard objectives from the literature [4], combinations of terms, and a handful of custom terms, 37 cost functions on the straight road and 39 on the merge road were tested.

5.5 Solver Robustness and Verification

IPOPT was limited to 300 iterations per solve, without a time limit. This was done as applying a time limit could affect the consistency of the solutions, depending on the load of the local machine. Each solve is warm-started from the previous plan, meaning that the solver uses the previous solution as its initial guess.

The controller was set up to keep the previous limits if IPOPT was unable to find a solution. This was required as 20–27% of solves failed to converge on the straight road scenario, with the range being dependent on the cost function used. On the merge road, 42–53% of solves failed, however almost all of these occurred at the end of the simulation, when congestion no longer needed to be controlled.

In order to evaluate if solver failure would drastically affect the cost function evaluations, alternative options were considered. However a stronger IPOPT setup (more iterations, an adaptive barrier strategy and a retry), which roughly halved the failure rate, was observed to not significantly alter the jam reduction of the cost function (likely due to the failures occurring at inconsequential periods of the simulation). Thus it was deemed acceptable to use IPOPT for the task, however failure rates were still logged.

6 Experimental Method

6.1 Simulation Scenarios

The controller setup was tested on two different SUMO road scenarios, which are summarised in Table 4. Both scenarios use the same vehicle types and VSL implementation, each having nine signs controlling 200 m segments, with a 3.6 km unsigned section at the start. The legal limits set for the road were imported from the provided SUMO framework; on the merge road, limits start at 84.6 km/h and move up by 1.8 km/h per section.

The merge road was the original scenario provided with the project. In it, two lanes merge into one, with demand set higher than what a single lane can carry, so that a queue forms at the merge section. The queue continues to form upstream as long as the demand continues, and is fixed positionally by the road design. Thus, this test verifies which cost functions are most effective for reducing stationary traffic shockwaves (in the case of a permanent bottleneck).

The straight road scenario was added to determine if the controller could manage a moving shockwave. Thus, the scenario has two lanes the entire way through and at 600 s through the simulation, one vehicle 300 m past the final sign is commanded to slow down to 15% of the speed limit for 240 s and then continues as normal. Inflow rates which were favourable for a travelling jam need to match the jam discharge rate, which was measured to be approximately 4700 veh/h, and thus inflow rates of 4600 and 4800 veh/h were tested.

Table 4: Simulation scenarios.

	Merge road	**Straight road**
Layout	3.6 km approach, 1.8 km signed, then a 0.5 km single-lane merge	3.6 km approach, 1.8 km signed, 1.2 km downstream; two lanes throughout
Legal limit	84.6 km/h on the approach, 84.6–99 km/h on the signs	84.6 km/h
Demand	4000 veh/h for 1000 s (about 164% of the merge's capacity)	4600 or 4800 veh/h for 2400 s (about 82% of capacity)
Cause of congestion	Permanent lane drop	One slow vehicle, 600–840 s
Without control	Standing queue reaching about 2.7 km upstream	Jam of 21 or 31 veh-h, lasting 660 or 956 s
Simulated time	3500 s	2700 s

This setup was found effective to produce a travelling jam, thus allowing for the testing of which cost functions are most effective for reducing travelling traffic shockwaves.

6.2 Performance Metrics

Several metrics were used for evaluating the cost functions, all of which were measured in SUMO, which acts as the ground truth.

Shockwave metrics: Vehicles were defined as being in a shockwave when measured to have a speed of less than 30 km/h. The primary metric used in evaluation was 'congestion', which is cumulative total time that all the road sections have spent in a shockwave, measured across the entire road. 'Jam vehicle-hours' (different to the jam term used in cost functions) was created from the total time the vehicles spend in the shockwave created by the slow vehicle (straight road only), which captures both the size of the jam and how much traffic encounters it. 'Front reach' was created by measuring how far upstream the merge road's queue extends at its peak. The vehicle-weighted acceleration [24] of vehicles was also considered as a metric; however, it was found to be reduced by simply slowing down traffic, rather than by reducing the shockwave.

Mobility metrics: Mobility was measured by total time spent (TTS), which included vehicles waiting at the entrance of the road [4]. It was reported as the mean change in total time per vehicle, compared with no control on the same seeds. The number of vehicles queued at the entrance was also checked, as it demonstrates if the controller is simply not allowing vehicles to enter the road in order to optimise for certain measures.

Safety metrics: SUMO provides several safety measures, such as time-to-collision (TTC, how many seconds till two cars would collide if they kept their current speeds), deceleration-rate-to-avoid-a-crash (DRAC [26], how drastically the vehicle would have to decelerate), and the vehicles speed. Several surrogate metrics were developed from these, including hard braking (a measure of where DRAC is above 3.35 m/s^2), as well as time-exposed-TTC (TET) and time integrated TTC (TIT), which measure how long and how severely cars are in a TTC of below 3 s. Hard braking was used as the main safety metric in analysis, as unlike TET and TIT, it does not require stopped vehicles to be filtered out.

6.3 Baselines and Statistical Protocol

In order to establish the effectiveness of each cost function, two baselines were used to compare to. The first was applying no control, with every sign displaying the legal limit for its segment. The second was a reactive rule, which represents typical current practice of slowing down upstream segments to a set speed once a jam is detected. The reactive rule was based off the Dutch automatic incident detection system [27] and the SPECIAL-IST algorithm [28]. The rule measures the average speed over 30 s intervals, and if any section drops below 30 km/h, the speed limits of the upstream sections are set to 40 km/h for at least 120 seconds. Thus, the rule is completely reactive, and performs no prediction. The rule was extensively tuned on the scenarios to ensure it represented a strong baseline of reactive techniques. Its detection threshold was tuned on seeds 1–10, and in testing it gave the same 81% jam reduction at 4600 veh/h on a further unseen 20 seeds, which shows that its performance is not specific to the seeds it was tuned on.

As SUMO randomises driver behaviour (reflecting how drivers are imperfect in reality), each configuration was run over 10 different seeds. The same seeds were used for each run to ensure consistency, and a Wilcoxon signed-rank test [29] was used for comparison. This test ranks seeds based off how consistently one setup beats another over all the seeds, in order to prevent a cost function being determined as superior by chance. The output of the test is p, where a smaller value correlates to a higher confidence in a difference between the two setups being compared (with $p < 0.05$ being assumed to be a significant confidence). With 10 seeds, the smallest possible p is 0.002, meaning every seed moved in the same direction.

6.4 Assumptions

Table 5: Main assumptions.

Assumption	**Effect on results**
Drivers obey limits instantly	Favours MPC
Exact traffic measurements will be available	Both controllers perform better than with real detectors
High SUMO capacity (2900 veh/h/lane)	The flow rates don't affect reality, however they are indicative of the road's capacity
Traffic demand is constant	Varying demand has not been tested
One layout and jam cause per road	Limited test-set
Surrogate safety measures are accurate	SUMO drivers cannot crash, so the simulation only accounts for near misses

One of the most important assumptions from Table 5 is that drivers follow speed limits instantly. This inherently favours MPC as MPC assumes this behaviour from drivers, however the traffic model used in MPC can also be adjusted to assume more realistic driver behaviour. Another key assumption which may point to results in practice being worse, is that the measurements of traffic are assumed to be perfect. In actuality, there is likely to be slight error, however the consequence of this was not measured. The main assumption which affects the comparison of cost functions, which runs underneath all the assumptions in Table 5, is that the traffic model used by the controller is accurate. The implications of this assumption will be further explored in Section 7.

7 Results and Discussion

7.1 Merge Road: Standing Queue

Figure 3 shows that no cost function was able to reduce congestion, and out of the 39 cost functions, 14 of them significantly increased congestion. The reactive rule was the only policy that lowered both congestion (−2.8%) and time (−4.0 s per vehicle), however neither of the changes were statistically significant (failing the Wilcoxon test). Table 6 shows the key results.

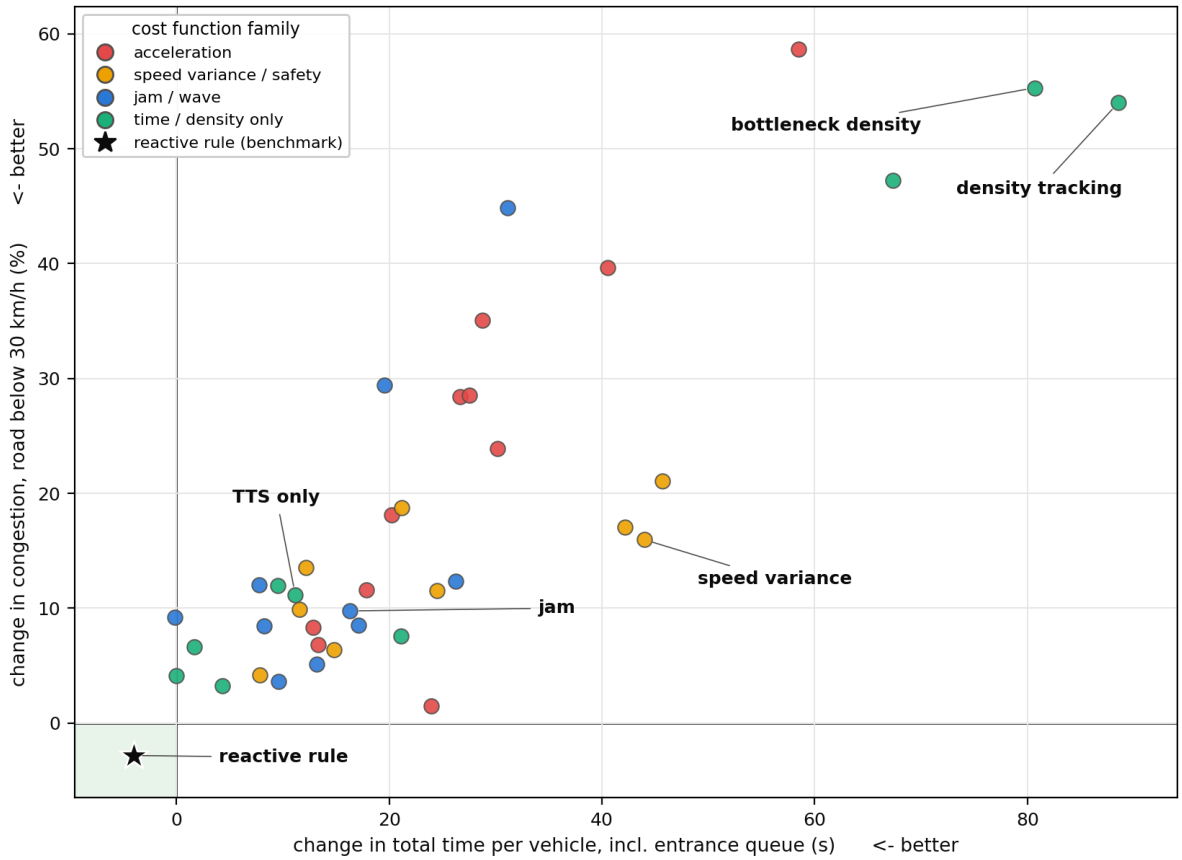


Figure 3: Merge road: congestion against time per vehicle for each policy, relative to no control.

The reason that no policy could improve congestion for the merge scenario was likely due to the intrinsic behaviour of a merge bottleneck. As the inflow was set higher than the discharge rate (approximately 4000 vs 2150 veh/h), shockwaves were guaranteed to occur. Furthermore, since the amount of vehicles in the shockwave was fixed by the excess demand, the only way to improve traffic conditions was to 'smooth-out' the shockwave over a longer stretch of road. Seemingly what occurred, is that congestion has been measured to be roughly the same if not higher for some of the cost functions even with the shockwave being 'smoothed-out'.

This is because although there was less-intense acceleration, it occurred over a larger area. This claim was consistent with the measured sharpest change in speed between

Table 6: Merge road, key results relative to no control.

Objective	Congestion	p	Time (s/veh)	Front reach
TTS only	+11.1%	0.160	+11.2	+4.4%
TTS + control variation [4]	+6.6%	0.275	+1.6	+1.5%
TTS + jam	+9.8%	0.322	+16.3	+5.2%
TTS + speed variance	+16.0%	0.084	+44.0	+7.4%
TTS + conflict risk	+9.9%	0.232	+11.6	+3.0%
TTS + upstream congestion	+5.1%	0.322	+13.2	−0.7%
Density tracking (preliminary design)	+54.0%	0.004	+88.5	+7.4%
Reactive rule	−2.8%	0.695	−4.0	+4.4%

road segments across runs. Compared to no-control (which has a sharpest drop of 45 km/h), the rule based method reduces it by 1.2%, whereas TTS (by itself, and combined with jam and conflict risk) reduces it by 5.1%, 5.2%, and 6.8% respectively. This can be seen in Figure 4, where the y axis has been replaced with sharpest-change in speed, compared to Figure 3 where it is congestion.

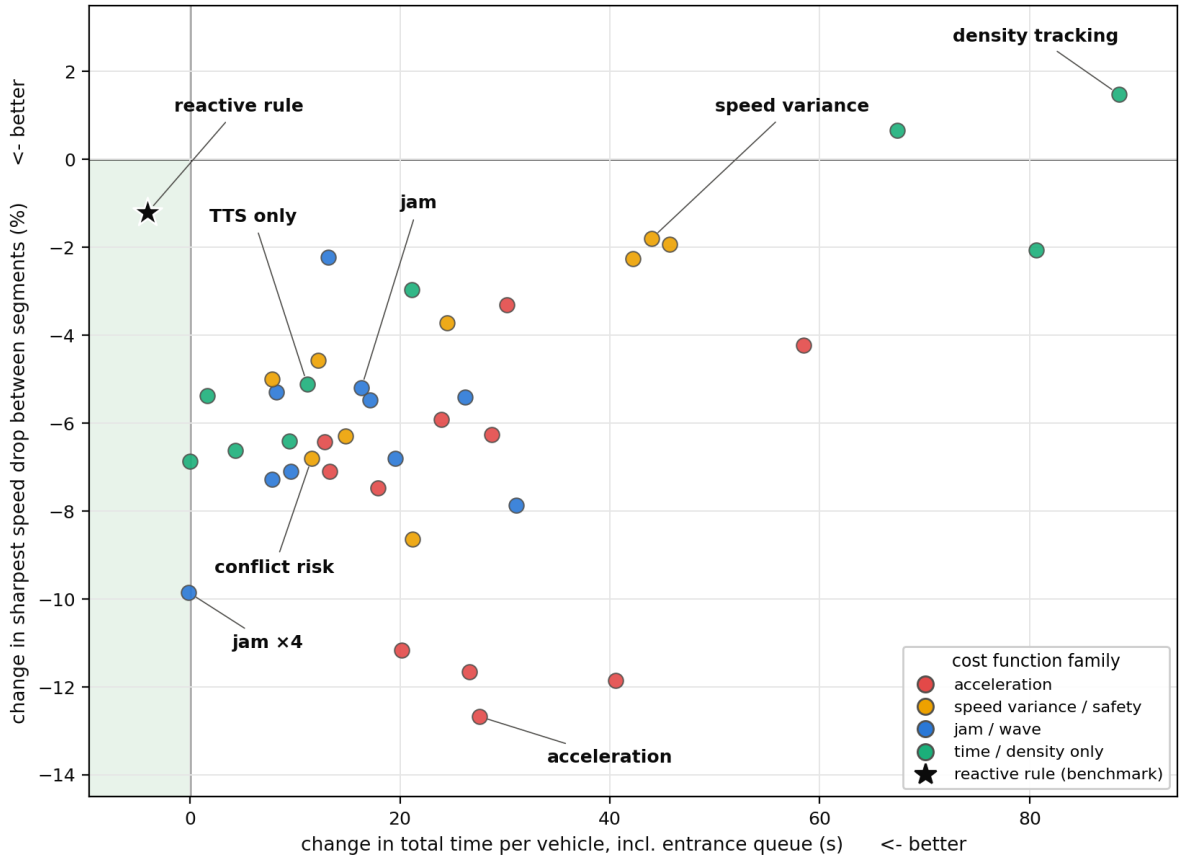


Figure 4: Merge road: sharpest-change in speed against time per vehicle for each policy, relative to no control.

7.2 Straight Road: Cost Function Comparison

When the congestion directly created by the slow vehicle was measured, every objective was found to reduce it, in the range of 50–85% (in jam vehicle-hours). However, when all congestion was measured (including the congestion created as an effect of the control law), only 15 of the 37 objectives at 4600 veh/h (seen in Figure 5), and 17 objectives at 4800 veh/h (seen in Figure 6) reduced it significantly. The objectives that increased congestion (which were mainly the acceleration, density-rate, and density-squared components) were found to shrink the slow vehicle's jam by holding traffic back, thus creating new congestion elsewhere. The numerical results can be seen in Table 7.

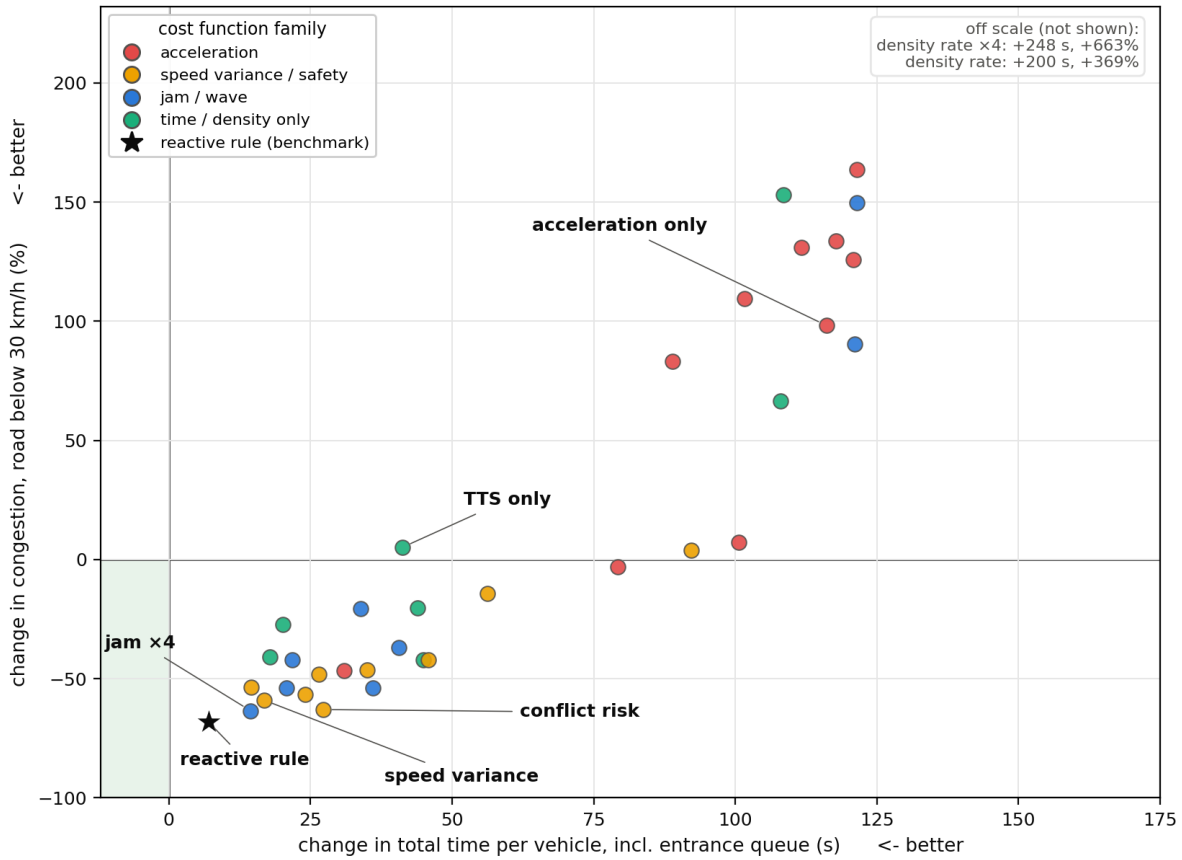


Figure 5: Straight road, 4600 veh/h: congestion against time per vehicle for each policy, relative to no control.

Unlike the merge scenario, several objectives were found to significantly reduce congestion, however no objectives were found which reduced congestion and reduced TTS. The best of these objectives were observed to be a combination of TTS with the jam term, conflict risk, or speed variance, reducing congestion by 58–71% for 14–61 s of extra time per vehicle. However, these cost functions did not perform better than the reactive rule, which was able to achieve a similar congestion reduction (68–70%) but with only 7–8 s of extra time added per vehicle (full numerical results can be seen in Table 7, and for every cost function in Tables 12 and 13).

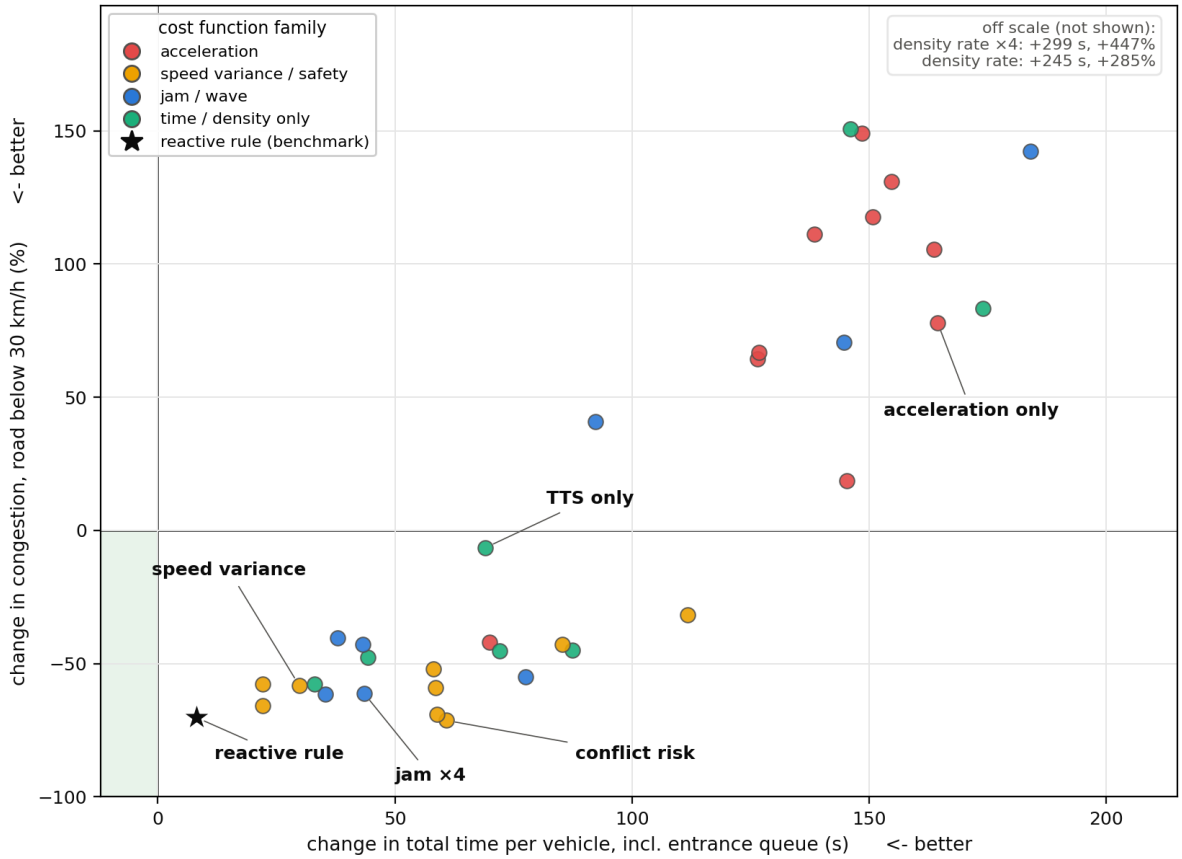


Figure 6: Straight road, 4800 veh/h: congestion against time per vehicle for each policy, relative to no control.

Table 7: Straight road, key objectives relative to no control (10 seeds). † is used to mark results which were not reliable across seeds ($p \geq 0.05$).

	Congestion		**Time (s/veh)**		**Sharpest drop**	
Objective	4600	4800	4600	4800	4600	4800
TTS only	+5%†	−6%†	+41	+69	+16%†	+10%†
TTS + control variation	−20%†	−45%†	+44	+72	+13%	+3%†
TTS + jam (×4)	−63%	−61%	+14	+44	+4%†	+8%†
TTS + speed variance	−59%	−58%	+17	+30	+4%†	+3%†
TTS + conflict risk	−63%	−71%	+27	+61	+9%	+8%†
Acceleration only	+98%†	+78%	+116	+164	+20%	+20%
TTS + density rate	+369%	+285%	+200	+245	+60%	+57%
Reactive rule	−68%	−70%	+7	+8†	−8%	−13%

The reactive rule reduced the sharpest speed drop better than the majority of the cost functions (8–13% reduction for the reactive rule, compared to a median increase of +24% among the 26 of the 37 cost functions that significantly increased it at 4600 veh/h). This was determined to likely be caused by the controller's ability to post different speed limits on neighbouring signs, whereas the reactive rule posts the same limit on every sign behind the jam. Full results for sharpest speed drop for the most prospective cost functions can be seen in Table 7, and why the controller performs worse than the rule is explored further in Section 7.5.

7.3 Safety

Further analysis was completed in the realm of safety for the reactive rule and five MPC cost functions from Section 7.2, for the straight road scenario, to allow for further differentiation between the objectives. Table 8 shows that every objective substantially reduces hard braking, by 57–72% for MPC objectives, and 77-78% for the rule. Time integrated time-to-collision (TIT) fell by 33–44% for three of the objectives, compared to 53–57% for the rule. However, only the rule was able to significantly and consistently reduce time exposed time-to-collision (TET) (at 27–34%). Hard breaking is a preferable metric of analysis compared to TET, as the latter removes vehicles which are stopped.

Table 8: Straight road: change in safety measures relative to no control.

	Hard braking		**TIT**		**TET**	
Policy	4600	4800	4600	4800	4600	4800
TTS + control variation	−57%	−60%	+6%	−10%	+52%	+58%
TTS + jam (×2)	−68%	−68%	−43%	−10%	−14%	+32%
TTS + jam (×4)	−65%	−68%	−44%	−33%	−17%	+6%
TTS + speed variance	−71%	−65%	−37%	−36%	−4%	−3%
TTS + conflict risk	−66%	−72%	−43%	−41%	−4%	+16%
Reactive rule	−78%	−77%	−53%	−57%	−27%	−34%

It can be seen from Table 8 that though the cost functions delivered good results, none of them were able to beat the reactive rule. When comparing the results to Table 7, it can be seen that there is a very high correlation between congestion reduction and TIT reduction ($r = 0.997$ at 4600 veh/h and 0.93 at 4800 veh/h for TIT, and 0.80 and 0.65 for hard braking), meaning that the policies which removed the most congestion also removed the most crash risk. The safety results are also consistent with the most valuable cost functions for reducing congestion without drastically increasing TTS being a combination of TTS with a shockwave or safety term, with the strongest appearing to be the jam term but also speed variance and conflict risk.

7.4 Realistic Sign, Driver Compliance and the Rule Benchmark

The results reported in the previous sections assume that drivers start to obey speed limits as soon as they are posted, however in reality this is not the case. Therefore, one cost function (TTS + jam (×2)) was run with realistic sign setup, with drivers travelling at 1.5 times the posted limit. This was used as an indicator of how much MPC and the reactive rule are affected by drivers not obeying the limit, and the 1.5 times is not meant

to be indicative of what would occur in reality.

Table 9: Straight road: MPC (TTS + jam ×2) and reactive rule under different sign and driver conditions, relative to no control.

	Congestion		**Time (s/veh)**	
Policy and condition	4600	4800	4600	4800
MPC, idealised sign	−68%	−77%	+31	+28
MPC, realistic sign	−39%	−65%	+25	+31
MPC, realistic sign, drivers 1.5×	−38%	−37%	+1	+1
Reactive rule	−68%	−70%	+7	+8
Reactive rule, drivers 1.5×	−60%	−60%	0	−1

It can be seen in Table 9 that with drivers travelling at 1.5 times the posted limit, MPC's congestion reduction fell to 37–38% on the realistic sign, whereas the rule fell only to 60%. In both cases the added time disappeared, as drivers were able to drive faster. Thus it can be seen that when drivers do not follow the limit exactly, the MPC model can degrade in its effectiveness, though this varied depending on the inflow.

The performance of MPC when used idealistically (no rounding of limits, running every 30 seconds) was seen to be close to, if not surpassing the reactive rule (at 68–77%, Table 9), although at about four times the added delay. This was likely partly due to MPC's traffic model not being aware of the speed limits being rounded, creating a mismatch between the model and its execution.

7.5 Why the Controller Behaves This Way

MPC is only able to choose effective speed limits if the model it uses is able to accurately predict the effects of those limits. This was seen as the main factor behind why MPC was unable to beat a reactive approach. The rounding of control actions may contribute to the model's inaccuracy (Section 7.4), however the model's ability to predict the outcomes that its control actions would cause was also flagged. This was tested by applying fixed sign limits throughout traffic simulations, and comparing what the macroscopic model believed would occur with the change SUMO produced. The results can be seen in Figure 7, and shows that the macroscopic model is rarely able to perfectly predict the effects of varying the speed limits of sections.

Figure 7 shows that on the straight road, the model predicted the traffic would move in the correct direction 72% of the time, but was unable to accurately predict the severity of the effect (−2.0 against −8.2 veh-h on average). The model was also seen to predict that the effect of the signs would be zero for several instances. These were measured to be in the gap before the disturbance was measured, and during the period when the model was not sure that the queue would keep building enough for it to warrant actions. If the model were to act in the few minutes where it was deciding, it would have a far greater effect. When measured in SUMO, if the model were to act at onset of the disturbance rather than 2 minutes later, it reduced the jam by up to 12% more. When these zero predictions are ignored, it can be seen that overall the model gets the direction, but not severity, correct for the straight road scenario.

Figure 7 shows that on the merge road scenario, the model does not predict the effects of altering speed limits well. For example, it predicted that a 40 km/h restriction would shrink the queue by 4–6 veh-h, when in SUMO it grew it by 4–15 veh-h. It should also be noted that for this test, a 420 s horizon was used, which is beyond the 210 s horizon used to validate the model against persistence (Table 2).

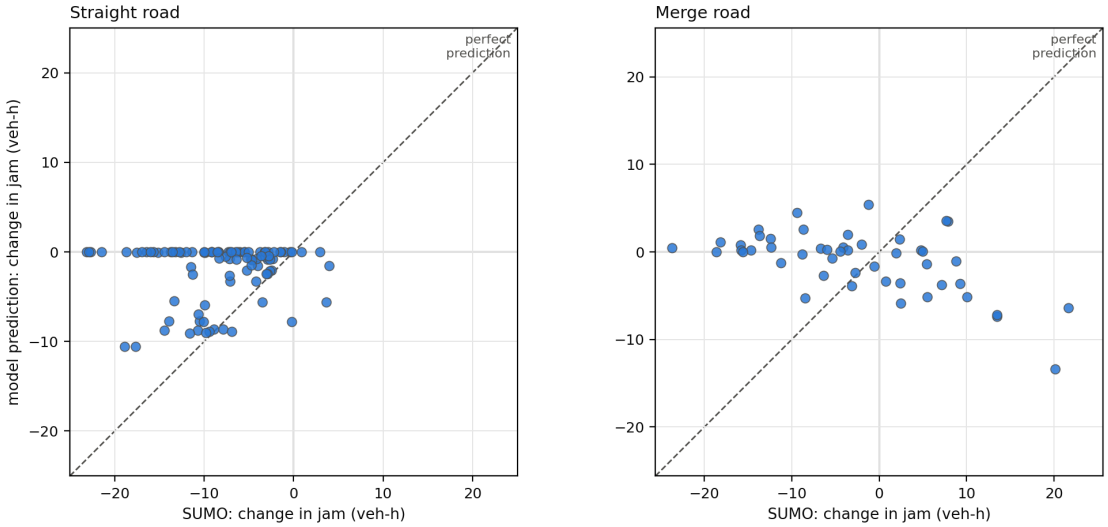


Figure 7: Change in jam predicted by the model against the change in SUMO, for each sign plan.

7.6 Discussion

The research question asked which MPC cost functions best reduce shockwaves on a highway, and at what cost to mobility and safety. Although the results and trends varied across the two scenarios tested, using TTS with a shockwave or safety term was found to give the best trade-off between reducing shockwaves and safety risk, and the loss of mobility.

On the merge road scenario no cost function was able to reduce both congestion and travel time. This was identified as a limit of the road due to the excess demand physically required to be stored somewhere. However, several cost functions were observed to reduce the sharpness of the speed change between sections more effectively than the reactive rule. Out of these cost functions, TTS with jam (x4) (Figure 4) was found to reduce sharpest-drop 9% more than the reactive rule, with no additional TTS, but with 9.2% more congestion compared to no control (though was found to be incosistent from the Wilcoxon test). Along with TTS + terminal (−6.9% sharpest-drop), it was one of only two cost functions in this scenario that decreased the severity of the shockwave without increasing TTS.

On the straight road scenario the selection of cost functions was seen to have a far more polarising effect on the results. The most effective cost functions were seen to be a combination of TTS and a shockwave or safety term that incorporates the shockwave directly (such as jam, conflict risk or speed variance). Across these three configurations (Tables 7 and 8), congestion was reduced by 58–71% and hard braking by 65–72%, for 14–61 s of

extra time per vehicle. The cost functions which did not reduce either TTS or congestion were those which aimed at smoothing the flow of traffic (such as acceleration or density rate), as well as cost functions which didn't incorporate a travel-time term. This was observed to be due to their holding back of vehicles, which shrank the slow vehicle's jam but created new congestion elsewhere on the road. In the pairings of TTS with safety terms, weighting the jam term more heavily was found, at 4600 veh/h, to yield both a larger shockwave reduction and produce less delay, although this did not hold at 4800 veh/h or for the speed variance and speed drop terms. This was seen to indicate that although TTS was required to ensure time-delay didn't increase to unreasonable measures, the safety terms were the most effective objectives.

The travel time and speed variance pair of objectives recommended by Hadiuzzaman et al. [3] and Yang et al. [6] were found to be effective on travelling shockwaves (once tuned), but ineffective on stationary shockwaves. Using standard travel time with control variation [4] was also found to be ineffective in reducing congestion. The view that MPC is far more effective than reactive control did not match the conclusions from the tests ran in this project, as a simple rule modelled from real life practices [27, 28] matched the best objectives on congestion, improved hard braking, and added only 7–8 s per vehicle compared to the best MPC cost function results.

MPC's inability to outperform a simple reactive rule was determined to be due to its inaccurate predictions of how changing speed limits would affect traffic, seen in Figure 7 and discussed in Section 7.5. Furthermore, using a realistic version of MPC and altering driver compliance was found to greatly reduce its effectiveness (Table 9) (as large of an effect as altering the cost function). This is likely partly due to the optimiser not knowing that its output would be rounded.

It appears that in practice, using a reactive based rule captures a large portion of the benefits MPC produces. MPC has shown to be able to reduce specific aspects of traffic flow (such as sharpest-drop), and yields the benefit of being more configurable. However, due to MPC requiring an accurate model, it is far harder to setup and test than reactive based rules. In all future testing, whole road congestion should be measured to avoid policies 'cheating' by slowing down traffic, or moving the congestion before the entrance point. All the conclusions made in this report were developed using simulations, with idealised drivers, on two scenarios (discussed further in Section 7.7).

7.7 Limitations

- **Idealised drivers and data:** It was assumed that SUMO's drivers follow speed limits as soon as they are posted, and that the controller operates on perfectly measured vehicle counts and speeds from each section. Both of these assumptions lead to artificially better results than what may occur with MPC in actuality.

- **Limited scenarios:** Due to a compute and time budget, only two scenarios were tested. The two most important aspects of shockwaves were explored (travelling vs stationary), however more scenarios may give additional insight into the varying performance of cost functions across alternate situations.

- **Prediction model:** The model was calibrated to match SUMO for a no-control

situation, rather than correctly predicting the effect of the VSL signs (identified as the main constraint of MPC in Section 7.5). The model also runs off different queue values compared to SUMO (approximately half to two-thirds), and is fundamentally unable to predict shockwaves before they appear in the travelling shockwave scenario, rendering its predictive capabilities useless in that regard.

- **Controller simplifications:** Predictions are made by a model which assumes continuous speed limits are applicable, whereas in actuality they are rounded before being posted. Unlike the safety constraint used by Hegyi et al. [8], no limit was placed on the difference between neighbouring signs. The realistic sign, which includes this rounding, was found to affect the results more than changing the cost function objectives, in some scenarios.

- **Measurement coverage:** Due to limited compute, safety metrics were only measured for five MPC objectives and the rule.

- **Statistical power and solver:** Only 10 seeds were tested per configuration, meaning minor effects cannot be validated before their results can be trusted. The solver failed 20–27% of solves on the straight road, and 42–53% of solves on the merge road, however the majority of these occurred after the shockwave had drained and were likely to not change the results.

8 Conclusion

This project aimed to find which MPC cost functions are best at reducing congestion and increasing safety, when used on a variable speed limit system. A nonlinear MPC model with a calibrated Bounded-METANET macroscopic traffic model were tested, using SUMO as the ground truth. 39 cost functions were tested on a merge scenario with a standing queue, and 37 were tested on a straight road with a moving jam.

In the merge scenario, no cost function was found to reduce both congestion and travel time, due to the bottleneck being stationary. However, TTS paired with the jam term (at four times its matched weight) was found to reduce sharp speed gradients effectively, while keeping travel time similar to the no-control baseline. For the moving shockwave, it was found that pairing travel time as an objective with a shockwave or safety term (such as the jam term) reduced congestion by 58–71% and hard braking by 65–72% (with the greatest improvements at 4600 veh/h occurring when the jam term was more heavily weighted). However, for the straight-road scenario no cost function was found that could drastically outperform the tuned reactive rule. This was deemed to be due to the model not fully understanding the effect on the flow of traffic that changing the speed limits would have.

Reactive methods are the most immediate and easily implementable option for road operators, however MPC is able to yield similar results while being far more complicated and thus presents further potential if the implementation is improved. MPC on the merge scenario also demonstrated the ability to reduce sharper speed gradients, more effectively than the reactive rule. Future work should be done on improving the model's ability to correctly predict the effect on traffic of changing the VSL signs, as well as giving the

model direct insight into how its outputs will be rounded to 10 km/h steps. For more confidence in the model's real life applicability, further testing with realistic driver response and sensor noise, over varied scenarios should be tested.

References

[1] H. Deng and K. Zhou. "Research on variable speed limit of highway main line based on model predictive control". In: *2022 IEEE 10th Joint International Information Technology and Artificial Intelligence Conference (ITAIC)* 10 (2022), pp. 2106–2111. DOI: 10.1109/ITAIC54216.2022.9836808.

[2] M. L. Delle Monache et al. "New frontiers of freeway traffic control and estimation". In: *2022 IEEE 61st Conference on Decision and Control (CDC)*. 2022, pp. 6910–6915. DOI: 10.1109/CDC51059.2022.9993221.

[3] M. Hadiuzzaman et al. "Evaluating performance of a proactive optimal variable speed limit control using different objective functions". In: *13th COTA International Conference of Transportation Professionals (CICTP 2013)*. Vol. 96. Procedia - Social and Behavioral Sciences, 2013, pp. 2895–2906. DOI: 10.1016/j.sbspro.2013.08.321.

[4] A. Hegyi et al. "Model predictive control for optimal coordination of ramp metering and variable speed limits". In: *Transportation Research Part C: Emerging Technologies* 13.3 (2005), pp. 185–209. DOI: 10.1016/j.trc.2004.08.001.

[5] P. Barbecho Bautista et al. "How does the traffic behavior change by using SUMO traffic generation tools". In: *Computer Communications* 181 (2021), pp. 1–13. DOI: 10.1016/j.comcom.2021.09.023.

[6] X. Yang et al. "Proactive optimal variable speed limit control for recurrently congested freeway bottlenecks". In: *Transportation Research Board 92nd Annual Meeting*. Paper #13-3139. 2013, pp. 1–20.

[7] M. Kontorinaki et al. "First-order traffic flow models incorporating capacity drop: Overview and real-data validation". In: *Transportation Research Part B: Methodological* 106 (2017), pp. 52–75. DOI: 10.1016/j.trb.2017.10.014.

[8] A. Hegyi et al. "Optimal Coordination of Variable Speed Limits to Suppress Shock Waves". In: *IEEE Transactions on Intelligent Transportation Systems* 6.1 (2005), pp. 102–112. DOI: 10.1109/TITS.2004.842408.

[9] S. Wang et al. "Optimal Control of Autonomous Vehicles for Traffic Smoothing". In: *IEEE Transactions on Intelligent Transportation Systems* 23.4 (2022), pp. 3842–3854. DOI: 10.1109/TITS.2021.3094552.

[10] S. Wang et al. "A general approach to smoothing nonlinear mixed traffic via control of autonomous vehicles". In: *Transportation Research Part C: Emerging Technologies* 146 (2023), p. 103967. DOI: 10.1016/j.trc.2022.103967.

[11] M. Shang et al. "Extending ramp metering control to mixed autonomy traffic flow with varying degrees of automation". In: *Transportation Research Part C: Emerging Technologies* 151 (2023), p. 104119. DOI: 10.1016/j.trc.2023.104119.

[12] M. Abdel-Aty et al. "Evaluation of variable speed limits for real-time freeway safety improvement". In: *Accident Analysis Prevention* 38.2 (2006), pp. 335–345. DOI: 10.1016/j.aap.2005.10.010.

[13] F. Wu et al. "Optimal driving for vehicle fuel economy under traffic speed uncertainty". In: *Transportation Research Part B: Methodological* 154 (2021), pp. 175–206. DOI: 10.1016/j.trb.2021.10.010.

[14] R. Yu and M. Abdel-Aty. "An optimal variable speed limits system to ameliorate traffic safety risk". In: *Transportation Research Part C: Emerging Technologies* 46 (2014), pp. 235–246. DOI: 10.1016/j.trc.2014.05.016.

[15] S. Siri et al. "Freeway traffic control: A survey". In: *Automatica* 130 (2021), p. 109655. DOI: 10.1016/j.automatica.2021.109655.

[16] F. van Wageningen-Kessels et al. "Genealogy of traffic flow models". In: *EURO Journal on Transportation and Logistics* 4.4 (2015), pp. 445–473. DOI: 10.1007/s13676-014-0045-5.

[17] A. Messmer and M. Papageorgiou. "METANET: A macroscopic simulation program for motorway networks". In: *Traffic Engineering and Control* 31.9 (1990), pp. 466–470.

[18] W. Zhao et al. "Bounded-METANET: A new discrete-time second-order macroscopic traffic flow model for bounded speed". In: *Transportation Research Part C: Emerging Technologies* 180 (2025), p. 105345. DOI: 10.1016/j.trc.2025.105345.

[19] F. Airaldi. *sym-metanet: Symbolic modelling of highway traffic networks with METANET*. Python package, version 1.1.2. 2024. URL: https://github.com/FilippoAiraldi/sym-metanet.

[20] J. A. E. Andersson et al. "CasADi: A software framework for nonlinear optimization and optimal control". In: *Mathematical Programming Computation* 11.1 (2019), pp. 1–36. DOI: 10.1007/s12532-018-0139-4.

[21] F. Airaldi. *csnlp: Nonlinear programming with CasADi*. Python package, version 1.6.8. 2024. URL: https://github.com/FilippoAiraldi/casadi-nlp.

[22] A. Wächter and L. T. Biegler. "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming". In: *Mathematical Programming* 106.1 (2006), pp. 25–57. DOI: 10.1007/s10107-004-0559-y.

[23] Y. Han et al. "Resolving freeway jam waves by discrete first-order model-based predictive control of variable speed limits". In: *Transportation Research Part C: Emerging Technologies* 77 (2017), pp. 405–420. DOI: 10.1016/j.trc.2017.02.009.

[24] S. K. Zegeye et al. "Integrated macroscopic traffic flow, emission, and fuel consumption model for control purposes". In: *Transportation Research Part C: Emerging Technologies* 31 (2013), pp. 158–171. DOI: 10.1016/j.trc.2013.01.002.

[25] A. Alessandri et al. "Nonlinear optimization for freeway control using variable-speed signaling". In: *IEEE Transactions on Vehicular Technology* 48.6 (1999). DOI: 10.1109/25.806796.

[26] D. F. Cooper and N. Ferguson. "Traffic studies at T-junctions. 2. A conflict simulation record". In: *Traffic Engineering & Control* 17.7 (1976), pp. 306–309.

[27] S. F. Varotto et al. "Driver speed compliance following automatic incident detection: Insights from a naturalistic driving study". In: *Accident Analysis & Prevention* 150 (2021), p. 105939. DOI: 10.1016/j.aap.2020.105939.

[28] A. Hegyi et al. "SPECIALIST: A dynamic speed limit control algorithm based on shock wave theory". In: *2008 11th International IEEE Conference on Intelligent Transportation Systems*. IEEE, 2008, pp. 827–832. DOI: 10.1109/ITSC.2008.4732611.

[29] F. Wilcoxon. “Individual comparisons by ranking methods”. In: *Biometrics Bulletin* 1.6 (1945), pp. 80–83. DOI: 10.2307/3001968.

A Full Cost Function Results

Tables 11–13 list every cost function, sorted by the change in congestion. All values are changes relative to no control on the same 10 seeds, with the realistic sign. p is the Wilcoxon signed-rank p-value for the congestion change; mean limit is the average posted limit (km/h); failures is the percentage of solves in which the solver did not converge. Weights are shown relative to each term's matched weight (Table 10); w marks a fixed weight.

Table 10: Matched weight of each term on each road (TTS has weight 1).

Term	**Straight road**	**Merge road**
Terminal	40	200
Jam	2.5	1.3
Acceleration	0.1	0.3
Temporal accel.	6	9
Density-weighted accel.	0.025	0.002
Density rate	30000	1200
Upstream congestion	3	0.09
Speed variance	1.2	0.6
Speed drop	60	30
Conflict risk	60	80
Squared density	2	0.7
Density tracking	4	3
Entrance queue	0.1	5
Control variation	1.5	0.55
Bottleneck density	–	40
Bottleneck discharge	–	0.0007

Table 11: Merge road: every cost function relative to no control.

Objective	**Congestion**	p	**Time (s/veh)**	**Sharpest drop**	**Mean limit**	**Solver failures**
TTS + Acceleration + Entrance queue	+1%	0.846	+24	−6%	83.4	47%
TTS + Bottleneck discharge	+3%	0.695	+4	−7%	84.3	45%
TTS + Jam + Control variation	+4%	0.625	+10	−7%	83.4	50%
TTS + Terminal	+4%	0.625	0	−7%	84.3	46%
TTS + Speed drop ($\times\frac{1}{4}$)	+4%	0.557	+8	−5%	83.8	47%
TTS + Upstream congestion	+5%	0.322	+13	−2%	84.6	42%
TTS + Jam + Speed drop	+6%	0.389	+15	−6%	82.9	46%
TTS + Control var. (standard)	+7%	0.275	+2	−5%	84.1	46%
TTS + Density-weighted accel.	+7%	0.375	+13	−7%	82.7	47%
Delay only	+8%	0.492	+21	−3%	83.7	45%
TTS + Temporal accel. ($\times\frac{1}{4}$)	+8%	0.193	+13	−6%	82.0	45%
TTS + Density rate ($w = 1$)	+8%	0.193	+8	−5%	84.0	45%
TTS + Jam ($\times\frac{1}{4}$)	+9%	0.275	+17	−5%	82.9	45%
TTS + Jam ($\times 4$)	+9%	0.160	0	−10%	83.7	47%
TTS + Jam	+10%	0.322	+16	−5%	83.8	47%
TTS + Conflict risk	+10%	0.232	+12	−7%	83.0	48%
TTS only	+11%	0.160	+11	−5%	83.5	46%
TTS + Speed variance ($\times\frac{1}{4}$)	+12%	0.160	+24	−4%	81.3	50%
TTS + Acceleration ($\times\frac{1}{4}$)	+12%	0.160	+18	−7%	80.6	49%
TTS + Squared density	+12%	0.193	+9	−6%	84.4	47%
TTS + Density rate ($\times\frac{1}{4}$)	+12%	0.131	+8	−7%	84.2	46%
Jam only	+12%	0.160	+26	−5%	81.2	49%
TTS + Speed drop	+14%	0.123	+12	−5%	82.7	44%
TTS + Speed variance	+16%	0.084	+44	−2%	81.4	47%
TTS + Speed variance ($\times 4$)	+17%	0.027	+42	−2%	81.4	47%
TTS + Jam + Acceleration	+18%	0.010	+20	−11%	80.2	47%
TTS + Speed drop ($\times 4$)	+19%	0.084	+21	−9%	82.3	49%
Speed var. + TTS + Control var.	+21%	0.033	+46	−2%	80.2	46%
TTS + Temporal accel.	+24%	0.027	+30	−3%	79.8	49%
Accel. + TTS + Queue + Control var.	+28%	0.010	+27	−12%	77.2	48%
TTS + Acceleration ($\times 4$)	+29%	0.010	+28	−13%	79.5	53%
TTS + Density rate	+29%	0.020	+20	−7%	81.6	46%
TTS + Acceleration	+35%	0.010	+29	−6%	79.1	52%
Acceleration only	+40%	0.002	+41	−12%	78.7	49%
TTS + Density rate ($\times 4$)	+45%	0.004	+31	−8%	79.6	49%
TTS + Density tracking	+47%	0.002	+67	+1%	80.3	44%
Density tracking + Control var.	+54%	0.004	+89	+1%	77.4	49%
TTS + Bottleneck density	+55%	0.002	+81	−2%	79.2	47%
TTS + Temporal accel. ($\times 4$)	+59%	0.002	+58	−4%	78.3	51%
Reactive rule	−3%	0.695	−4	−1%	88.3	0%

Table 12: Straight road, 4600 veh/h: every cost function relative to no control.

Objective	**Congestion**	p	**Time (s/veh)**	**Sharpest drop**	**Mean limit**	**Solver failures**
TTS + Jam (×4)	−63%	0.002	+14	+4%	75.6	21%
TTS + Conflict risk	−63%	0.002	+27	+9%	72.1	23%
TTS + Speed variance	−59%	0.002	+17	+4%	77.1	21%
TTS + Speed variance (×4)	−57%	0.002	+24	+12%	76.7	21%
TTS + Jam	−54%	0.004	+21	+6%	75.4	21%
TTS + Jam + Control variation	−54%	0.002	+36	+9%	69.5	22%
TTS + Speed variance ($\times\frac{1}{4}$)	−54%	0.002	+15	+7%	77.9	23%
Speed var. + TTS + Control var.	−48%	0.002	+26	+10%	73.2	22%
TTS + Temporal accel. ($\times\frac{1}{4}$)	−47%	0.002	+31	+11%	72.9	21%
TTS + Jam + Speed drop	−46%	0.020	+35	+14%	71.6	26%
TTS + Speed drop ($\times\frac{1}{4}$)	−42%	0.006	+46	+19%	72.2	22%
TTS + Density tracking	−42%	0.010	+45	+16%	69.5	24%
TTS + Jam ($\times\frac{1}{4}$)	−42%	0.027	+22	+5%	75.4	21%
TTS + Terminal	−41%	0.006	+18	+2%	76.4	23%
TTS + Upstream congestion	−37%	0.010	+41	+17%	73.8	20%
Delay only	−27%	0.084	+20	+4%	76.2	22%
TTS + Density rate ($w = 1$)	−21%	0.232	+34	+16%	74.5	21%
TTS + Control var. (standard)	−20%	0.375	+44	+13%	70.0	20%
TTS + Speed drop	−14%	0.432	+56	+20%	70.5	21%
TTS + Temporal accel.	−3%	0.896	+79	+34%	68.0	23%
TTS + Speed drop (×4)	+4%	0.865	+92	+31%	68.9	26%
TTS only	+5%	0.770	+41	+16%	74.3	21%
Accel. + TTS + Queue + Control var.	+7%	0.922	+101	+18%	63.2	26%
Density tracking + Control var.	+66%	0.232	+108	+42%	62.9	23%
TTS + Acceleration + Entrance queue	+83%	0.193	+89	+24%	68.0	25%
TTS + Density rate ($\times\frac{1}{4}$)	+90%	0.010	+121	+47%	67.0	26%
Acceleration only	+98%	0.131	+116	+20%	62.6	24%
TTS + Jam + Acceleration	+109%	0.027	+102	+25%	67.4	26%
TTS + Acceleration (×4)	+126%	0.010	+121	+22%	63.8	26%
TTS + Density-weighted accel.	+131%	0.010	+112	+28%	67.0	24%
TTS + Acceleration ($\times\frac{1}{4}$)	+134%	0.004	+118	+39%	66.5	25%
Jam only	+150%	0.027	+121	+36%	63.9	23%
TTS + Squared density	+153%	0.002	+108	+62%	71.9	23%
TTS + Acceleration	+163%	0.002	+121	+32%	66.0	25%
TTS + Temporal accel. (×4)	+214%	0.002	+163	+79%	62.5	23%
TTS + Density rate	+369%	0.002	+200	+60%	60.9	25%
TTS + Density rate (×4)	+663%	0.002	+248	+88%	59.9	22%
Reactive rule	−68%	0.002	+7	−8%	81.2	0%

Table 13: Straight road, 4800 veh/h: every cost function relative to no control.

Objective	**Congestion**	p	**Time (s/veh)**	**Sharpest drop**	**Mean limit**	**Solver failures**
TTS + Conflict risk	−71%	0.002	+61	+8%	72.2	25%
TTS + Jam + Speed drop	−69%	0.002	+59	+1%	72.2	23%
TTS + Speed variance ($\times\frac{1}{4}$)	−66%	0.002	+22	−4%	77.2	21%
TTS + Jam	−62%	0.002	+35	−1%	75.2	21%
TTS + Jam (×4)	−61%	0.002	+44	+8%	74.7	22%
Speed var. + TTS + Control var.	−59%	0.002	+59	+2%	71.5	22%
TTS + Speed variance	−58%	0.004	+30	+3%	77.3	21%
TTS + Speed variance (×4)	−58%	0.002	+22	−3%	77.2	21%
TTS + Terminal	−58%	0.002	+33	+3%	76.1	23%
TTS + Jam + Control variation	−55%	0.002	+77	+10%	69.9	22%
TTS + Speed drop ($\times\frac{1}{4}$)	−52%	0.004	+58	+8%	72.7	22%
Delay only	−48%	0.014	+44	+7%	75.8	21%
TTS + Control var. (standard)	−45%	0.084	+72	+3%	68.7	25%
TTS + Density tracking	−45%	0.006	+87	+13%	68.7	24%
TTS + Upstream congestion	−43%	0.020	+43	+9%	74.6	23%
TTS + Speed drop	−43%	0.020	+85	+15%	70.6	23%
TTS + Temporal accel. ($\times\frac{1}{4}$)	−42%	0.037	+70	+15%	72.8	22%
TTS + Jam ($\times\frac{1}{4}$)	−40%	0.027	+38	0%	75.3	21%
TTS + Speed drop (×4)	−32%	0.105	+112	+22%	69.3	25%
TTS only	−6%	0.770	+69	+10%	73.4	22%
Accel. + TTS + Queue + Control var.	+19%	0.750	+145	+18%	62.7	24%
TTS + Density rate ($w = 1$)	+41%	0.432	+92	+21%	72.0	22%
TTS + Temporal accel.	+64%	0.131	+127	+31%	68.0	24%
TTS + Jam + Acceleration	+67%	0.064	+127	+19%	67.7	26%
TTS + Density rate ($\times\frac{1}{4}$)	+71%	0.275	+145	+33%	66.9	24%
Acceleration only	+78%	0.037	+164	+20%	62.5	24%
Density tracking + Control var.	+83%	0.027	+174	+39%	62.9	25%
TTS + Acceleration (×4)	+106%	0.004	+164	+23%	63.5	25%
TTS + Acceleration	+111%	0.020	+138	+20%	66.8	26%
TTS + Acceleration + Entrance queue	+118%	0.004	+151	+23%	66.1	27%
TTS + Density-weighted accel.	+131%	0.010	+155	+25%	66.3	26%
Jam only	+142%	0.004	+184	+38%	63.4	24%
TTS + Acceleration ($\times\frac{1}{4}$)	+149%	0.006	+149	+30%	67.4	25%
TTS + Squared density	+151%	0.004	+146	+47%	71.7	22%
TTS + Temporal accel. (×4)	+178%	0.002	+194	+57%	61.6	23%
TTS + Density rate	+285%	0.002	+245	+57%	62.3	24%
TTS + Density rate (×4)	+447%	0.002	+299	+74%	59.9	22%
Reactive rule	−70%	0.002	+8	−13%	81.6	0%